

PUSHKAR CHATURVEDI

Software Engineer | Backend Systems

☎ +91-9993205227 — ✉ pushkarchaturvedi42@gmail.com — 🌐 PushkarOM — 🔗 LinkedIn

Education

Jaypee University of Engineering and Technology

Aug 2023 – Present

B.Tech, Computer Science and Engineering

CGPA: 9.4/10

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Computer Networks, DBMS, Machine Learning, Probability & Statistics

Certifications: Andrew Ng Machine Learning Specialization — Mathematics for Machine Learning

Experience

Data & Backend Engineering Intern

May 2025 – Jul 2025

Inventlix (Remote)

- Engineered production RESTful APIs for an inventory management platform using Django REST Framework, with pagination, filtering, and sorting for secure data access.
- Designed and implemented JWT authentication, Google OAuth, and role-based access control (RBAC), embedding role claims directly into tokens to streamline authorization across production APIs.
- Integrated PostgreSQL, Google Cloud Storage, and SendGrid to support persistent data storage, media uploads, and automated transactional email workflows.
- Collaborated with backend engineers in a Git-based development workflow to ship production-ready features, maintaining modular architecture and code quality across the codebase.

Projects

RepoSage — Agentic RAG Codebase Assistant 🌐

Ongoing

Python, FastAPI, LangChain, Chroma, Docker, AWS

- Architected an agentic Retrieval-Augmented Generation (RAG) platform that indexes GitHub repositories for semantic code search and repository understanding, using FastAPI, LangChain, and Chroma.
- Implemented tool-enabled retrieval workflows, conversational memory, and file-level search to help developers navigate large, unfamiliar codebases.
- Automated backend validation with PyTest and GitHub Actions, running the test suite on every code change.
- Containerized the application with Docker and deployed it to AWS EC2, producing a reproducible build and deployment workflow.

SkillSwap — Community Learning Platform 🌐

Ongoing

React, Node.js, Express, MongoDB, JWT

- Co-developing a full-stack MERN platform with a peer engineer, enabling collaborative learning through shared study sessions, reflections, and community interactions.
- Designed a modular Express backend with JWT access/refresh authentication, email verification, and rate limiting to secure REST APIs.
- Modeled scalable MongoDB schemas and integrated backend APIs with the React frontend; currently adding Docker-based CI/CD for cloud deployment.

EmotionSense — Multimodal Emotion Assistant 🌐

Ongoing

FastAPI, React, PyTorch, Hugging Face

- Developed a modular FastAPI backend serving real-time multimodal emotion inference across speech and text models.
- Integrated the inference pipeline with an LLM-powered conversational layer, emphasizing modular architecture and scalable backend design.

Technical Skills

Languages: Python, Java, C++, JavaScript, SQL, C

Backend & APIs: Django REST Framework, FastAPI, Node.js, Express.js, REST APIs, JWT, OAuth, Celery

Databases: PostgreSQL, MongoDB, Redis

Cloud, DevOps & Testing: Docker, AWS EC2, GitHub Actions (CI/CD), PyTest, Google Cloud Storage, Linux

Frontend: React.js, Tailwind CSS

AI/ML: LangChain, RAG, Chroma, FAISS, PyTorch, Hugging Face, Scikit-learn

Tools: Git, Pandas, NumPy, Matplotlib

Leadership & Achievements

Secretary, ROS Division (ROSPINOT) | Organizing Team, CodeSrijan Hackathon | Top 25 Finalist, Bharat AI SoC Student Challenge | GATE CS 2026: AIR 7514 (Score: 500)